

Pure Maths 1.1 · Quadratics

discriminants

Simultaneous equations

Completing the square

$$ax^2 + bx + c \rightarrow a(x - k)^2 + h$$

(h, k) is the vertex

if $a \neq 1$, factor out a :

$$= a(x^2 + \frac{b}{a}x) + c$$

$$= a\left(x + \frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 + c$$

$$= a(x - k)^2 + h$$

Example:

$$y = 2x^2 + 8x + 5$$

$$2(x^2 + 4x) + 5$$

$$2\left(x + \frac{4}{2}\right)^2 - \left(\frac{4}{2}\right)^2 + 5$$

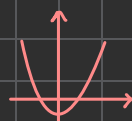
$$2(x + 2)^2 - 8 + 5$$

$$= 2(x + 2)^2 - 3 \#$$

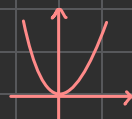
$[-2, -3]$: Vertex

discriminant of $ax^2 + bx + c = 0$ is given by: $[b^2 - 4ac]$

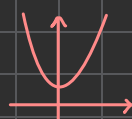
if $b^2 - 4ac$ is



> 0 , two real unequal roots



$= 0$, one real equal root



< 0 , no roots, not real

Eg: $y = x - 5$ — ①

$5y + x^2 + 10 = -21$ — ②

Substitute eq. ① into eq. ②.

Eg:

$$5(x - 5) + x^2 + 10 = -21$$

~ now solve for x

$$5x - 25 + x^2 + 10 = -21$$

$$x^2 + 5x + 6 = 0$$

$$(x + 2)(x + 3) = 0$$

$$x = -3, x = -2$$

Sub x into ① to find y .

Solving Quadratic Equations

Factorizing

Eg: $x^2 + 5x - 6$

look for the sum of factors of -6 that equals to the b value, 5.

therefore, roots are $x = -6$ & $x = 1$

$$[6x - 1 \rightarrow 6 - 1 = 5]$$

Quadratic Formula

$$ax^2 + bx + c$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Substitute values of a, b, c into formula to find the roots.

Calculator

fx-570ES:

[mode] + [5, EQN]

+ [3, $ax^2 + bx + c$]

press values of a, b, c & don't forget negative sign if exists.

Recognizing Quadratic in other forms & solving.

equations like

$$x^4 + 5x^2 + 6 = 0$$

Can be written as a quadratic if you let $x^2 = y$, and then:

$$(x^2)^2 + 5x^2 + 6 = 0$$

$$y^2 + 5y + 6 = 0$$

Solve for y then, $y = x^2$, find x .

try eq. like $\tan^2 \theta + \tan \theta + 2 = 0$ in same.

Pure Maths 1.2 : Functions



Key Concepts & definitions

Function: A rule that assigns each element from the domain (input values) to exactly one element in the range (output values).

Domain: the set of all possible input values (x -values) for a function.

Range: the set of all possible output values (y -values) for a function.

One-One function: a function where each input has a unique output, different x values give diff y -values. This is necessary for function to have inverse.

Inverse function: A function that reverses the effect of original function, denoted as $f^{-1}(x)$. for the function $f(x)$, $f(f^{-1}(x)) = x$ & $f^{-1}(f(x)) = x$.

Composite function: Combination of two functions, denoted as $gf(x)$ where function f is applied first to x , the output of $f(x)$ is applied into $g(x)$.

Example: $f(x) = x + 3$ $fg(x)$ — g into $f(x)$ so, $fg(x) = (x^2 + x) + 3$
 $g(x) = x^2 + x$ find value of $fg(3)$ — $\rightarrow fg(3) = (3)^2 + 3 + 3 = 15 \#$

Identifying Domain and Range

Domain: The set values for all x -values for which function is defined, domain can't be 0 if x is denominator of a function or negative in Square root.

Range: range of all possible output values eg: for $y = \frac{1}{x-1}$, $y \neq 0$ as $0 = \frac{1}{x-1}$ means $\frac{0}{0} = x-1$ undefined.

Composite of function:

a composite function $gf(x)$ is only defined when range of $f(x)$ lies within domain of $g(x)$.

Inverse function range & domain

for inverse functions:

range of $f(x)$ = domain of $f^{-1}(x)$

domain of $f(x)$ = range of $f^{-1}(x)$.

Common domain & ranges

• $\sin \theta$, $\cos \theta$, range $\rightarrow 1 \geq f(x) \geq -1$

• $\tan \theta$, domain $x \neq 90$ or $\frac{\pi}{2}$ / odd values of $90^\circ / \frac{\pi}{2}$.

One-One & Inverse functions

One-One functions

to determine if a function is one-one, check if diff x values always give diff y values. Eg:
 $f(x) = x^2$ is not one-one cause $f(-2) = f(2)$.
 but $f(x) = x^3$ is one-one.

How to find Inverse of a function

Example: Find inverse of $f(x) = \left(\frac{x-3}{5}\right) \times 6$

#1: Swap the x for y , rearrange eq. to make x subject.

$$x = \left(\frac{y-3}{5}\right) \times 6$$

$$\frac{x}{6} = \frac{y-3}{5} \quad \text{therefore, the inverse } f^{-1}(x) \text{ is}$$

$$5\left(\frac{x}{6}\right) = y - 3$$

$$5\left(\frac{x}{6}\right) + 3 = y$$

$$f^{-1}(x) = 5\left(\frac{x}{6}\right) + 3$$

Graph Transformations

Vertical translation

$y = f(x) + a$, shift up if $a > 0$, down if $a < 0$.

Horizontal Translation

$y = f(x+a)$ shift left if $a > 0$, right if $a < 0$.

Vertical Stretch

$y = af(x)$, stretch factor a

Horizontal Stretch

$y = f(ax)$, stretch factor $\frac{1}{a}$

Reflection

In x -axis: $-f(x)$

In y -axis: $f(-x)$

Combined Transformations

Multiple transformations can be applied together:

Example: $y = 2f(x+3) - 1$

the graph $y = f(x)$ has been

#1: horizontally translated 3 units to left

#2: Vertically stretched by factor 2.

#3: Vertically translated 1 unit down.

* Order of translations:

horizontal shift first, then horizontal stretch, compression, reflect then, Vertical stretch, compression, reflect and lastly vertical shift.

Pure Maths 1.3 : Coordinate Geometry!

Equation of a Straight Line

Forms of Line Equations

Slope Intercept form: $y = mx + c$ — m : gradient, c : y -intercept

point-slope form: $y - y_1 = m(x - x_1)$ — m : gradient

general form: $ax + by + c = 0$

Parallel and Perpendicular Lines

• lines are parallel if their gradient is same.

• lines are perpendicular if the product of their gradients is -1 .

↳ if you have gradient of one line and need to find gradient of its perpendicular
 Use the formula $-\frac{1}{m_1}$ where m_1 is gradient of original line, $-\frac{1}{m_1}$ is gradient of perpendicular.

Perpendicular bisector of line

$-\frac{1}{m_1}$ = gradient

↳ intersect at midpoint of original line.

#1: Find Perpendicular gradient

#2: Find midpoint

#3: Use gradient & midpoint in $y - y_1 = m(x - x_1)$ to find line eq. of Perpendicular bisector.

to form the equation of a straight line you need 2 main things: gradient m and y -intercept c .

$$\text{Gradient } (m) = \frac{y_2 - y_1}{x_2 - x_1}$$

to find y -intercept (c): $y - y_1 = m(x - x_1)$
 where y and x remain variable and y_1, x_1 is a set of coordinate in that line.

Other Formulas:

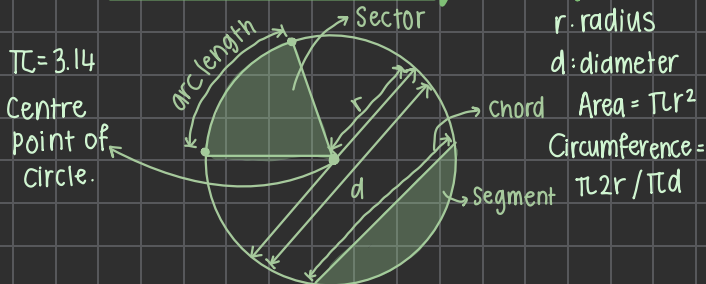
distance between 2 coordinates

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint of two coordinates

$$= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Circles and Their Equations



Equation of circle

$$(x-a)^2 + (y-b)^2 = r^2 \quad \text{where } (a,b) \text{ is centre}$$

r is radius

Note: tangent to circle is perpendicular to radius at point of contact.
 angle in semicircle is always 90° eg: always 90° .

Graphs, Intersections and Solving Equations

Intersection points of a line and a curve correspond to the solutions of the simultaneous equations of line and the curve.

Use discriminant ($b^2 - 4ac$) to determine how many times a line intersects curve:

$b^2 - 4ac = 0$: tangent (intersects at one point only)

$b^2 - 4ac > 0$: two intersection points

$b^2 - 4ac < 0$: never intersects.

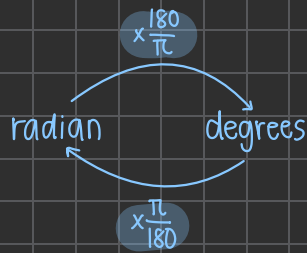
Eg: for $y = x + k$ find number of intersections with given curve by equating them, simplify and use discriminant..

Pure Maths 1.4: Circular Measure.

Arc length, Sector Area & etc.

Conversion of radian to degrees and back

a radian is a unit of angular measure, one radian is the angle opposite to center of circle by an arc whose length is equal to the radius of the circle.



or use ratio

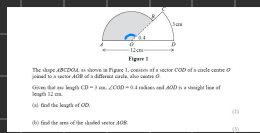
$$\pi : 180^\circ$$

Arc length (s)

$$s = r\theta$$

arc length $\leftarrow$ s , radius $\leftarrow$ r , angle in radian $\leftarrow$ θ

Example:



a) arc length CD is 3 cm
 θ is 0.4 rad

$$s = r\theta, 3 = r(0.4)$$

$$r = 7.5 \text{ cm } OD = 7.5 \text{ cm}$$

b) Area Sector: $\frac{1}{2}r^2\theta$
 $r = 7.5, \theta = 0.4$

$$\text{Area} = \frac{1}{2}(7.5)^2(0.4) = 11.25 \text{ cm}^2$$

Area of a Sector

$$A = \frac{1}{2}r^2\theta$$

radius $\leftarrow$ r , angle in radian $\leftarrow$ θ

Pure Maths 1.5: Trigonometry.

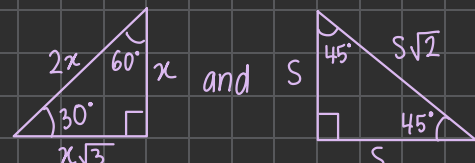
Exact Values of Angles

$$\sin 30^\circ = \frac{1}{2}, \cos 30^\circ = \frac{\sqrt{3}}{2}, \tan 30^\circ = \frac{1}{\sqrt{3}}$$

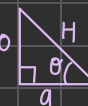
$$\sin 45^\circ = \frac{1}{\sqrt{2}}, \cos 45^\circ = \frac{1}{\sqrt{2}}, \tan 45^\circ = 1$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \cos 60^\circ = \frac{1}{2}, \tan 60^\circ = \sqrt{3}$$

Using Special Right Triangles



Note:



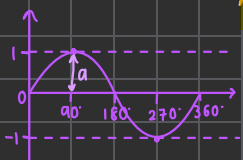
$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} \quad \text{SOH}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} \quad \text{CAH}$$

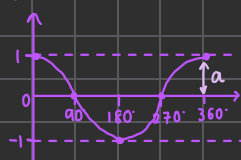
$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} \quad \text{TOA}$$

Graph of $\sin x$, $\cos x$ and $\tan x$

#1: $y = a \sin bx + c$



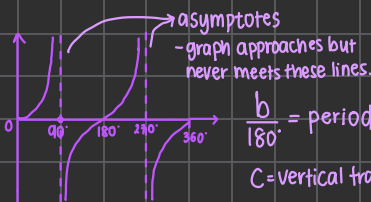
#2: $y = a \cos bx + c$



a: amplitude
b: period
c: vertical translation

#3: $y = a \tan bx + c$

no amplitude because it can go up till infinity.



asymptotes
- graph approaches but never meets these lines.
b: period
c: vertical translation
a: vertical stretch

Trigonometric Identities and Solving Equations

① $\sin^2 \theta + \cos^2 \theta = 1$

② $\frac{\sin \theta}{\cos \theta} = \tan \theta$

③ $\frac{1}{\sin \theta} = \text{cosec } \theta, \frac{1}{\cos \theta} = \text{sec } \theta, \frac{1}{\tan \theta} = \text{cot } \theta$

Using the Identities

Example: $\sec \theta - \sin \theta \tan \theta = \cos \theta$

$$\text{LHS} = \frac{1}{\cos \theta} - \sin \theta \left(\frac{\sin \theta}{\cos \theta} \right)$$

$$= \frac{1}{\cos \theta} - \frac{\sin^2 \theta}{\cos \theta}$$

$$= \frac{1 - \sin^2 \theta}{\cos \theta}$$

$$= \frac{\cos^2 \theta}{\cos \theta} = \cos \theta = \text{RHS, hence, Proven.}$$

Note:
try changing everything to sin & cos to make simplifying easier!

S $180^\circ - x$ Sin positive only	A all positive x
T $180^\circ + x$ tan positive only	C $360^\circ - x$ cos positive only

For a given range
Example $0 \leq x \leq 360$
u can find all possible values of a trigonometric function using ASTC. Example:

$$5 \tan^2 x = -16$$

$$\tan^2 x = -\frac{16}{5}$$

$$x^2 = \tan^{-1} \left(-\frac{16}{5} \right)$$

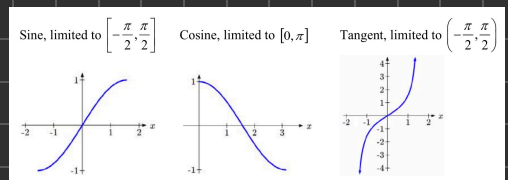
$$x = 72.64$$

angle, quadrant where T negative: C&S

Inverse Trigonometric Function

Inverse of $\sin \theta$, $\tan \theta$, $\cos \theta$ is $\sin^{-1} \theta$, $\cos^{-1} \theta$, and $\tan^{-1} \theta$, which gives the Principle values of angles for specific intervals. Example:

$\sin^{-1} x$ outputs angle between -90° and 90°
 $\cos^{-1} x$ outputs angle between 0° and 180°
 $\tan^{-1} x$ outputs angle between -90° and 90°



graphs of Inverse trig functions.

Pure Maths 1.6 : Series

Binomial Expansion

For the expansion of $(a+b)^n$ where n is a positive Integer: use the binomial theorem.

$$(a+b)^n = a^n + {}^nC_1 (a)^{n-1} (b)^1 + {}^nC_2 (a)^{n-2} (b)^2 + \dots$$

Until ${}^nC_{n-1} (a) (b)^{n-1} + b^n$.

Where nC_r is $\binom{n}{r}$ and is $= \frac{n!}{r!(n-r)!}$ Combination formula

Example: $(x+3)^5$

$$= x^5 + {}^5C_1 (x)^4 (3)^1 + {}^5C_2 (x)^3 (3)^2 + {}^5C_3 (x)^2 (3)^3 + {}^5C_4 (x) (3)^4 + (3)^5 \leftarrow \text{expand/simplify.}$$

* note: Press [Shift] + [=] for C in calc.

Geometric Progressions

a Sequence where each term is a constant multiple of the previous term. Example:

2, 4, 8, 16, 32, 64, 128

$\times 2$ $\times 2$ $\times 2$ $\times 2$

where $r=2$
 $a=2$
 first term in Progression.

Formulas

#1: $T_n = ar^{n-1}$ — n^{th} term formula

#2: $S_n = \frac{a(1-r^n)}{1-r}$ — Sum of n terms when $|r| \neq 1, r \neq 1$

#3: $S_n = \frac{a(r^n-1)}{r-1}$ — Sum of n terms when $r > 1, r < 1$

#4: $S_{\infty} = \frac{a}{1-r}$ — Sum to infinity only if $|r| < 1$

Arithmetic Progressions

a Sequence where the difference between consecutive terms is constant.

Example:

2, 4, 6, 8, 10

$+2$ $+2$ $+2$ $+2$

d (common difference) = 2

a (first term) = 2

Formulas

#1: $T_n = a + (n-1)d$ — n^{th} term

#2: $S_n = \frac{n}{2}(a+l)$ — Sum of n terms and l is last term.

#3: $S_n = \frac{n}{2}(2a + (n-1)d)$ — Sum of n terms.

Pure Maths 1.7 : Differentiation

Differentiation Rules

* Power Rule: $\frac{d}{dx} (x^n) = nx^{n-1}$

* Chain Rule: $\frac{d}{dx} (a(x^n-1)^b) = ab(x^n-1)^{b-1} (nx^{n-1})$

* Product Rule: $\frac{d}{dx} (u(x)v(x)) = u(x)v'(x) + v(x)u'(x)$

* Quotient Rule: $\frac{d}{dx} \left(\frac{u(x)}{v(x)} \right) = \frac{u'(x)v(x) - u(x)v'(x)}{v(x)^2}$

NOTE: Learn indices Rules to help —

$a^m \times a^n = a^{m+n}$ $a^0 = 1$ $a^{\frac{m}{n}} = \sqrt[n]{a^m}$
 $a^m \div a^n = a^{m-n}$ $a^1 = a$ $a^{-\frac{m}{n}} = \frac{1}{\sqrt[n]{a^m}}$
 $(a^m)^n = a^{m \times n}$ $a^{-m} = \frac{1}{a^m}$

NOTE:

$u'(x)$ is derivative of $u(x)$

$v'(x)$ is derivative of $v(x)$

Gradient of a Curve

* $\frac{dy}{dx}$ represents the gradient of the curve $y = f(x)$

* if value of $\frac{dy}{dx}$ at point (x_1, y_1) is m :

→ equation of tangent at that point is $y - y_1 = m(x - x_1)$

→ Equation of normal at that point is $y - y_1 = -\frac{1}{m}(x - x_1)$

Second derivatives

$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2y}{dx^2}$ or $f'(f'(x)) = f''(x)$

Example: $\frac{d^2}{dx^2} (x^4 + x^2 + 3)$

$\frac{dy}{dx} = 4x^3 + 2x$

$\frac{d^2y}{dx^2} = 12x^2 + 2$ #

Increasing and decreasing functions

$y = f(x)$ is increasing for given Interval of x if $\frac{dy}{dx} > 0$ throughout the Interval.

$y = f(x)$ is decreasing for given Interval of x if $\frac{dy}{dx} < 0$ throughout the Interval.

Connected Rates of Change

When two variables, x and y , both vary with a third variable, t , the three variables can be connected using the chain rule:

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

and NOTE:

$$\frac{dx}{dy} = \frac{1}{\left(\frac{dy}{dx} \right)}$$

The volume of a spherical balloon is increasing at a rate of $0.05 \text{ m}^3/\text{s}$.

How fast is the surface area increasing when the radius is 0.4 m ?

v = volume, s = surface area, r = radius

$v = \frac{4}{3}\pi r^3$ $s = 4\pi r^2$

$\frac{dv}{dr} = 4\pi r^2$ $\frac{ds}{dr} = 8\pi r$ $\frac{dv}{dt} = 0.05$

$\frac{ds}{dt} = \frac{ds}{dr} \times \frac{dr}{dt} \times \frac{dr}{dt}$

$= 8\pi r \times \frac{1}{4\pi r^2} \times 0.05$

$= \frac{0.1}{r}$

When $r = 0.4$, $\frac{ds}{dt} = \frac{0.1}{0.4} = 0.25$

Surface area is increasing at $0.25 \text{ m}^2/\text{s}$.

Stationary, Minimum and Maximum Points

Stationary (turning points) of a function $y = f(x)$

occurs when $\frac{dy}{dx} = 0$

At max/min point, $\frac{dy}{dx} = 0$ and the gradient is positive to left of max point and negative to right. Gradient is negative to left of min point and positive to the right.

If $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0$ it is a maximum point

If $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} > 0$ it is a minimum point.

If $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} = 0$, nature of stationary point can be found by

Pure Maths 1.8) Integration

Reverse differentiation

* Reverse process of differentiation is Integration
To Integrate $(ax+b)^n$ use:

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

Integral Symbol with respect to x Integral constant

Example:

$$\int (2x^2 + 5x) dx = \frac{2}{3}x^3 + \frac{5}{2}x^2 + C$$

$$\int (2x+1)^2 + 4 dx = \frac{\frac{1}{3}(2x+1)^3}{2} + 4x + C$$

$$= \frac{1}{6}(2x+1)^3 + 4x + C$$

* ALWAYS write Integral constant for Indefinite Integrals.

Evaluating Integral Constant

* solving for C using given Conditions:

- Integrate $\frac{dy}{dx}$ function to find eq. of original line, Substitute any known (x,y) coordinates to find C.

EXAMPLE:

$\frac{dy}{dx} = 2x+1$ and curve passes (1,-2)
find original curve equation.

$$y = \int (2x+1) dx$$

$$y = \frac{2}{2}x^2 + 1x + C$$

$$y = x^2 + x + C \quad \text{Substitute } (1,-2)$$

$$-2 = (1)^2 + (1) + C$$

$$C = -4 \quad \text{so: } y = x^2 + x - 4 \quad \#$$

Definite Integrals

The Integral with limits [a,b]

Formula:

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F(x) is integral of f(x) without the constant [+C].

Example:

$$\int_1^2 x^2 dx$$

$$= \left[\frac{1}{3}x^3 \right]_1^2$$

$$= \frac{1}{3}(2)^3 - \frac{1}{3}(1)^2$$

$$= \frac{7}{3} \quad \#$$

Area using Definite Integration

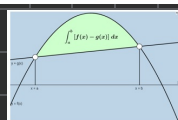
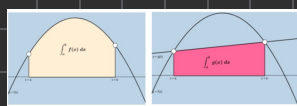
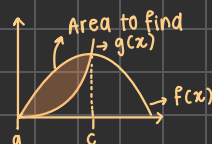
* to find Area under curve, between curve or with respect to lines

↳ Integrate eq. of curve and any other line

↳ Subtract lower curve from upper if needed.

Formula:

$$\text{Area} = \int_a^c [f(x) - g(x)] dx \quad \text{where:}$$



Example:

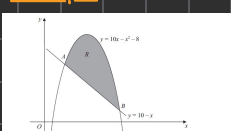


Figure 2 shows the line with equation $y = 10 - x$ and the curve with equation $y = 10x - x^2 - 8$. The line and the curve intersect at the points A and B, and O is the origin.

(a) Calculate the coordinates of A and the coordinates of B.

The shaded area R is bounded by the line and the curve, as shown in Figure 2.

(b) Calculate the exact area of R.

a) A & B are Intersection

Points of line $y = 10 - x$
with curve $y = 10x - x^2 - 8$

$$\text{So, } 10 - x = 10x - x^2 - 8$$

$$x^2 - 11x + 18 = 0$$

$$(x-9)(x-2) = 0$$

$$\begin{matrix} x=9 & x=2 & A: (2, 8) \\ y=10-x & y=10-x & B: (9, 1) \end{matrix}$$

b) for Area R:

$$\int_2^9 (10x - x^2 - 8) dx \quad \text{gives Area:}$$

we don't need area under line $10 - x$

$$\text{So: Area R} = \int_2^9 [10x - x^2 - 8] - \int_2^9 [10 - x]$$

$$\int_2^9 \left[\frac{11x^2}{2} - \frac{x^3}{3} - 18x \right] = \frac{81}{2} - \left(-\frac{50}{3} \right) = \frac{343}{6} \text{ unit}^2 \quad \#$$

Volumes of Revolution

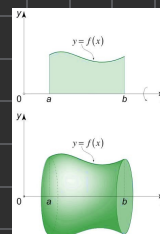
Volume generated by rotating a curve about an axis (y or x).

For rotation about x axis:

$$V = \pi \int_a^b [f(x)]^2 dx$$

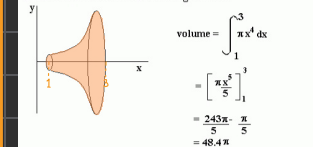
For rotation about y axis:

$$V = \pi \int_a^b [g(y)]^2 dy$$

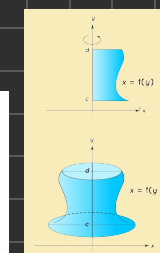


Example:

The graph of $y = x^2$ between $x = 1$ and $x = 3$ is rotated completely around the x-axis. Find the volume generated.



$$\begin{aligned} \text{volume} &= \int_1^3 \pi x^4 dx \\ &= \left[\frac{\pi x^5}{5} \right]_1^3 \\ &= \frac{243\pi}{5} - \frac{\pi}{5} \\ &= \frac{242\pi}{5} \end{aligned}$$



extra notes: